GPS Devices

Lost? Find yourself...

GPS — a buzzword that has everyone hopping around in anticipation and some governments in fear. Global Positioning System (GPS) is a technology that allows you to get information about your exact position on Earth, in terms of latitude, longitude and altitude. GPS receivers access information about your position from satellite links. These receivers are available for purchase for civilians and have now become an important part of any travellers' arsenal. GPS is a tool that is invaluable people who are moving through unfamiliar territory.

For example, picture yourself driving interstate between two cities. There is always the odd chance of you taking a wrong turn, or a damaged signboard making your life difficult. Traditionally you would be totally at the mercy of the local folk, if you happen to find any while driving through rural India, in particular. In case you do not get any help, you would be largely stranded, since cellphone coverage would also be absent. With a GPS unit, however, you would be in a position actually to figure out where you are with an accuracy of five metres or so. Cellphones that have GPS inbuilt work similarly to GPS devices. These usually have some sort of interface inbuilt on a standard platform like Symbian, Windows Mobile or some patented firmware. When tracking is turned on, the GPS device will receive data from satellites and this data is used to triangulate an approximate position. Maps are inbuilt into such devices and by triangulating its coordinates you can actually figure out where exactly you are. GPS is mostly a free service and you need to pay no fee to use it. However, map providers may charge you a fee to download and use their maps, although some maps are available for free, or there may be a subscription scheme. For example, Nokia maps are free for use with Nokia GPS phones — but there is a restriction on maps of any five cities in India. Assuming you need the maps of two extra cities you will need to pay to download them.

You can either buy a small GPS dongle which connects via Bluetooth to your cellphone or PDA, or buy a full-fledged receiver which has a screen itself and requires no other device to connect. In case you have a Bluetooth GPS dongle and a cellphone you will need to download a third party GPS software to your phone, such as GPS Tuner or GPS Watch, among others. The GPS receivers will have their own software inbuilt. Some systems are available for installation in cars and although these are costly as of now, and the handheld devices are more common, many people with cars are showing interest in such systems.

So what else can you do with GPS other than navigation to ensure you stay on track? Well besides tracking your whereabouts GPS can also be used to track your car.

Installing a GPS unit in your car will give you details about where your car has been and how much distance was covered. You could also get a real-time update on the whereabouts of your car at the moment, but your car would also need a transmitting unit. With such a setup it's possible to get your car back from thieves — if you've just bought a shiny new BMW, you could consider this. Some GPS devices-cum-transmitters are also available in very compact sizes meaning you could use them to keep an eye on your kids. Adventurers also have an alternate use for this technology. If you go driving or trekking to a remote region you can use a GPS device to map the path you take. Any GPS device will record the route you use. This route can be downloaded and software is available which will allow you to make your own map of the region with your trail marked out. This can be used to help other people who wish to use the same route; your recorded path can be a sort of map for them. GPS has even become popular for businesses involved with any sort of transport operations. Real time vehicle tracking is an important tool. In fact, almost any industry benefiting from precise location-based information will greatly benefit from GPS.

For me, I'd never been a user of GPS till one of our test centre guys went to Taiwan and brought home a small receiver unit. It was fun to use and surprisingly accurate, as I found out later when I used it to travel by road to Goa, that being the first time I was making the trip. I was surprised to see the mapping of GPS in our country being accurate. In fact I could make a direct comparison between GPS Tuner which is a popular GPS mapping software and Nokia Maps (Nokia's GPS software) and am pleased to report that both of these were accurate enough to get me out of a tight spot — although GPS Tuner was much more accurate with more landmarks recorded.

For those who do not want to spend on a costly GPS unit you have the option of buying just the receiver unit, which is usually as large as a pen drive. Besides the GPS receiver you need GPS maps. Some people may be concerned about the number of channels that the receiver has since this is directly related to the accuracy of your position. Do not be. Even if your device has just 16 channels, remember that the number of GPS satellites orbiting the earth is not more 24, and the chances that more than a handful of them are overhead of you simultaneously is very slim. For your information, just two satellites will give you your position in 2D. Four satellite fixes are sufficient to get your GPS position in 3D, i.e. including altitude. The Holux M1000 is one such USB/Bluetooth based GPS receiver. It has 32 channels and uses the MTK GPS engine. This device doesn't have its own screen and you will need a notebook or PDA to use it. It's available for Rs 5,000. The Holux M-241 goes a little further. It has the same USB and Bluetooth connect, but includes a small screen and actually looks like a roll of camera film. Besides a receiver, it also has the facility to log data and can display information such as altitude, latitude and longitude and speed. It also has a photo based geo-tagging system by which you can connect places you've been with snapshots making it easier to archive and access later. It can store 1,30,000 locations. With 32 channels and a battery life rated at 12 hours, the M-241 is available for Rs 9,000. The ultra Holux GP Slim is also available. This one is really small, like a tiny USB drive and is available for Rs 6,200. Its specifications are nearly identical to the M-241. All these systems are very compact, and can be easily pocketed. Note that these devices do not have inbuilt maps, and you will need to buy these separately. Destinator's SatGuide is one of the paid-for software around with detailed maps for 30+ important cities. There are 24 categories of points of interest including hotels and ATMs. It also has some special features where it blacklists roads that are damaged, or dangerous as "roads to avoid". This software is available for Rs 4,300. MapmyIndia iNav is priced at Rs 4,550. This one is solely for cellphones or PDAs and you get an SD card pre-loaded with MapmyIndia maps. Voice assistance is also provided, as is route re-calculation in case you take a wrong turn. Although this may seem like a Nokia Maps clone you do not need a GPRS (data) connection to use it. In case your phone doesn't have GPS inbuilt you will need a Bluetooth GPS dongle. Navizon, 3D Tracking, and G-Map Track are some software available for cellphones. Navizon has Google Maps support.

If you are looking at a complete GPS solution in the form of a GPS handheld device with the software preloaded then you need to shell out a bit more — however, since the support is paid as well, the maps tend to be more accurate than the free ones. Paid software is also available with GPS device bundles. Mio Technologies' PND (personal navigation device, it has no other name!) works with SatGuide's software. With a 3.5-inch touchscreen and SiRF III receiver, the PND is based on Windows CE 5.0 and has 512 MB of flash storage and 64 MB of RAM. It supports USB charging and plays back MP3s as well. The kit costs Rs 15,500 and it is installable in your car using a vacuum cup — this allows it to sit on your dashboard. MapmyIndia Navigator 2.0 is priced at Rs 18,000. It has a 3.5 inch screen, and supports voice and graphical map guidance, along with tags for points of interest. It uses the SiRFStar III GPS chipset and is powered by a 400 MHz Samsung processor. SatGuide's P360 is available for Rs 19,000. Its WM 6.0 based, with 1 GB of ROM and 64 MB of RAM; the rest of the hardware is similar to the Navigator 2.0. This one is mainly to be used as a car GPS device and this fact is emphasised by the relatively poor battery life which is only around 4.5 hours. All SatGuide systems carry a lifetime free mapping system so you can download updated maps for free. Most others have support for up to a year. Delphi's NAV 200 is available for Rs 18,999. It's a well built device with a 3.5-inch screen with 32 MB of inbuilt memory. It comes with MMI Navigations' GPS map preinstalled. Voxtel's Carrera X350 is yet another PND I came across. Priced at Rs 14,999 this was one of the smartest looking devices owing to its slim dimensions. It's got a 3.5-inch screen. The sad thing is its abysmal battery life — only an hour or so, which means you had better not keep it very far from your car's charger.

For me, I'd rather use either my Imate KJam or my N95 8GB along with a Bluetooth dongle, since these give me greater flexibility, not to mention better battery life. I'd go with the M-241 for Rs 9,000 — mainly because of its great battery life, and try one of the examples of free GPS software in the beginning. If you must use paid-for software then the iNav MapmyIndia looks like the most attractive option available.